

SENG 41333 — COMPUTER BASED OPERATIONS MANAGEMENT

Operations Management Analysis

A Study of 3axislabs — Software Product Engineering Company, Jaffna, Sri Lanka

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SE/2021/002

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What We Will Cover Today

- 1 Organization Overview** *M. Holins*
- 2 Core Operational Functions** *M. Holins*
- 3 Nature of Operations & Workflow** *S. Dilakshan*
- 4 Digitalization & Technology Stack** *S. Dilakshan*
- 5 The Retention Problem & Proposed Solution** *K. Nirojan*
- 6 Solution Benefits, Implementation & Recommendations** *S. Tharshana*

Who is 3axislabs?

- Software / product development company founded in 2020, headquartered in Jaffna, Sri Lanka.
- Positioning: “Empowering Businesses with Technology” — a technology partner from idea to deployment and maintenance.
- Currently 15+ employees and 25+ delivered projects/products.
- Vision: become a leader in innovative software solutions that transform industries globally.
- Mission: empower businesses through tailored, modern technology solutions for long-term growth.

FOUNDING TEAM



Prasanth Subendran

CEO & Co-Founder — 9+ yrs



Jestan N. Jeyapalan

Co-CTO & Co-Founder — 14+ yrs



Saahithyan Vigneswaran

Co-CTO & Co-Founder — 9+ yrs

A Hybrid Services + Product Company

Industry & Sector

IT / Software Product Engineering — serving international clients across Australia, Canada, Finland, Singapore, and Sri Lanka.

Core Services

AI App Dev

API Integration

UI / UX Design

Full-Stack Dev

Mobile Dev

Cloud & DevOps

OWN PRODUCTS

MindMap AI

800K+

active users
(company-published)

AI mind-mapping tool — 5M+ maps created, 29+ languages, 4.8 avg. rating.

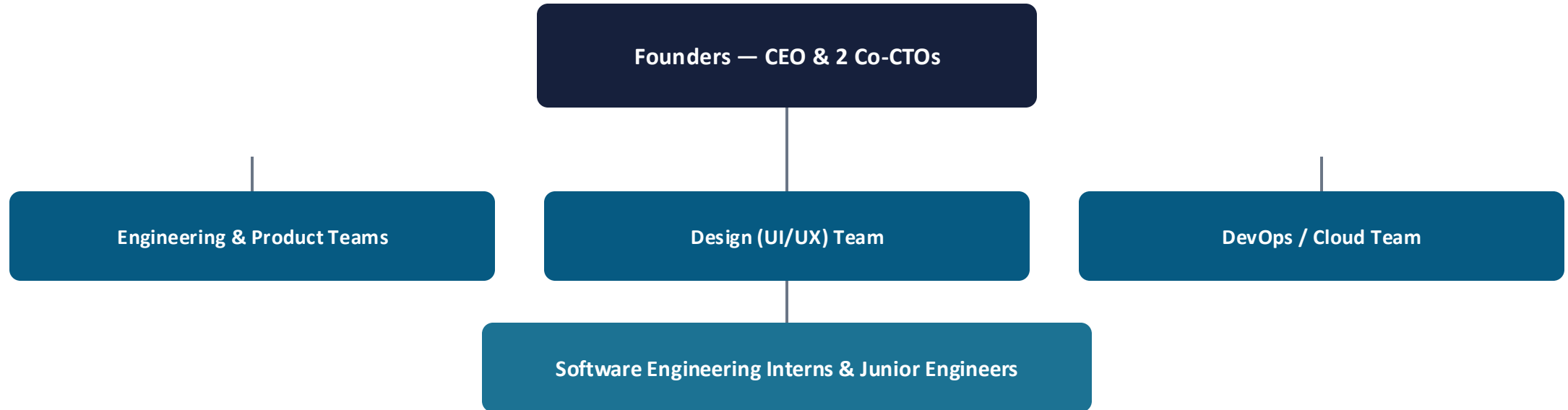
InternHub

Connects students, institutions & companies for internship placements. MVP delivered in ~3 months.

BusSwift

Customizable bus-transport platform with passenger/operator apps, live tracking & payments.

A Flat, Delivery-Focused Team Structure



- On-site Jaffna team; interns feed directly into live client and product delivery teams — a strength for hands-on learning, but a risk point (explained shortly).
- For clients, 3axislabs forms a “dedicated team” per engagement rather than a rigid departmental hierarchy.

A Hybrid Service & Product Operation



Service Operation

Custom software built to order for external clients — e.g., iSHIFT (real-estate CRM), Sharon Express (bus ticketing), Berry (sales insights).



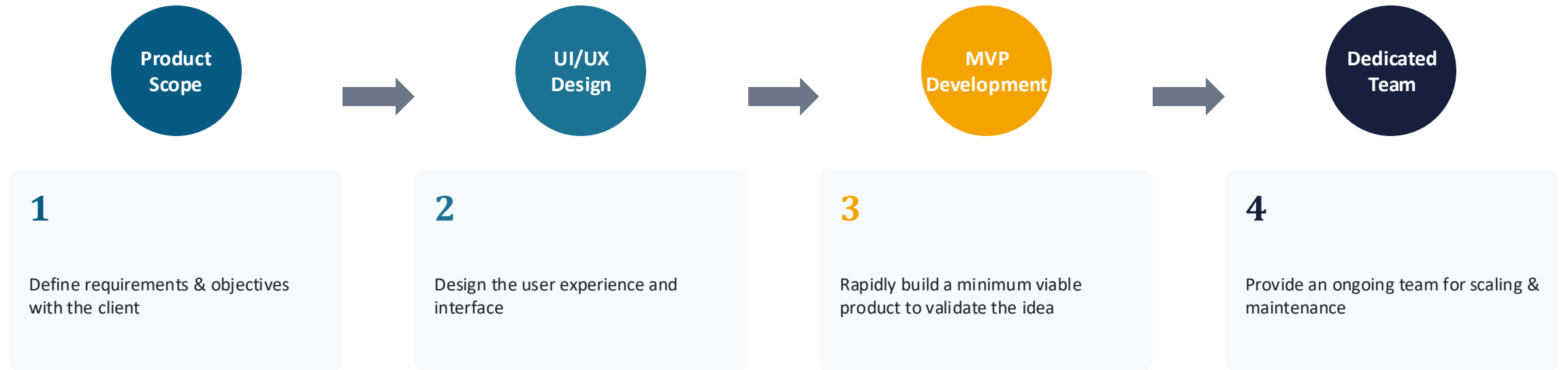
Product Operation

Company-owned SaaS products developed, hosted and continuously improved in-house — MindMap AI, InternHub, BusSwift.

Process Characteristics

- Project-based and highly customized — each engagement scopes unique requirements (Product Scope → UI/UX → MVP → Dedicated Team).
- Knowledge-intensive and people-dependent: value is created almost entirely through skilled developers, designers and engineers rather than physical assets.
- Iterative / agile delivery cycles with continuous testing, deployment and maintenance rather than a single hand-off.

The 3axislabs Product Development Pipeline



- This published process shows why 3axislabs favors startups and founders: it takes an idea from concept to a working, fundable MVP quickly, then scales delivery with a dedicated team.

Inputs → Transformation → Outputs



What Slows 3axislabs Down

2

Scaling with a small team

Only 15+ staff must cover 25+ projects/products across many industries.

3

Client work vs. product work

Balancing billable client delivery against maintaining owned products like MindMap AI.

4

Competing internationally

Winning and retaining clients in Australia, Canada, Finland etc. from a regional base.

1

THE #1 CHALLENGE

Talent & Intern Retention

Interns and junior engineers gain high-value, in-demand skills (Java, Spring Boot, React, AI tooling) during their time at 3axislabs — then leave for higher-salary offers at larger companies once they are market-ready.

This is the problem our proposed software solution addresses →

Systems Mapped to the Operations Framework

1

Information Systems

Cloud-based dev environments, Git version control & AI copilots (ChatGPT, Gemini) support daily engineering work.

2

ERP (Enterprise Resource Planning)

No dedicated in-house ERP platform is publicly disclosed — as an SME, resource planning likely runs on lightweight PM tools rather than a full ERP suite.

3

CRM (Customer Relationship Mgmt)

Integrates CRM platforms such as HubSpot for client-facing work, and builds CRM systems for clients (e.g., iSHIFT for real estate).

4

Inventory Management

Not directly applicable — 3axislabs delivers software, not physical goods; its closest equivalent is code repositories & cloud resource management.

5

Automation Technologies

Strong DevOps automation via GitHub Actions, Terraform & Ansible for CI/CD and infrastructure-as-code.

6

Data Analytics & Reporting

Uses Elasticsearch, Kafka & Spark for data processing, search and reporting pipelines across products.

DIGITALIZATION GAP

- Strong engineering, cloud, CRM & AI tooling for building client and product software.
- No dedicated internal HR-tech system for people operations, distinct from the CRM/ERP tools used for client work.
- No structured system to track intern skill growth, career pathways, or attrition risk.
- No systematic way to capture institutional knowledge before someone exits.

The Talent & Intern Retention Crisis

Interns are trained on real, live projects (Java, Spring Boot, React, cloud & AI tooling). Within months they hold market-ready, high-demand skills — and are then hired away by larger companies offering significantly higher salaries than an SME can match.

1

Repeated Training Cost

Every exit means re-investing months of onboarding & mentorship into a new hire.

2

Delivery Delays

Projects like InternHub's 3-month MVP sprints depend on continuity of trained staff.

3

Knowledge Loss

Tacit project knowledge leaves with the person — rarely documented beforehand.

4

Low Training ROI

The company effectively subsidizes larger competitors' talent pipelines.

Why Is This Happening?

1

Compensation Gap

As an SME, 3axislabs cannot always match the salary bands of large, well-funded companies.

2

No Visible Growth Path

Beyond the internship, there is little structured, visible career progression to justify staying.

3

No Attrition Data

There is no engagement or performance data to flag who is at risk of leaving before it happens.

4

Undocumented Knowledge

Tacit know-how lives in people's heads, not in a shared knowledge base, until they exit.

5

No Retention Incentives

No structured bonus, bonding, or milestone-based reward system tied to tenure or growth.

Introducing “TalentTrack”

A Talent Growth & Retention Management System for 3axislabs

1

Track Skill Growth

Give both intern and company clear, transparent visibility of skills gained over time.

2

Show Career Pathways

Make promotion routes and salary progression visible so growth doesn't require leaving.

3

Predict Attrition Risk

Use engagement & performance data to flag flight risk early, while it's still fixable.

4

Preserve Knowledge

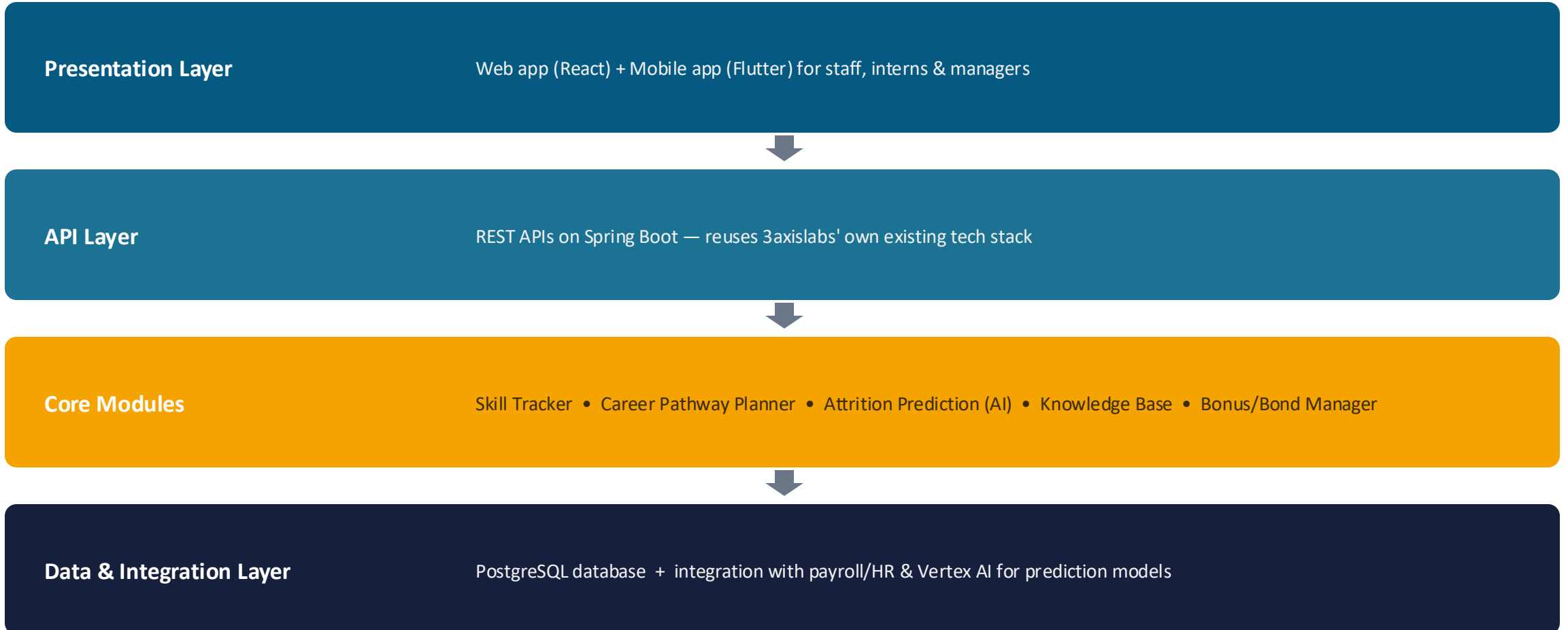
Capture project knowledge continuously, not only when someone hands in notice.

5

Automate Incentives

Manage retention bonuses & training-bond milestones systematically, not manually.

TalentTrack — High-Level Design



What TalentTrack Actually Does

1

Skill & Competency Dashboard

Visualizes each intern's growing skillset against role benchmarks.

2

Career Pathway Planner

Shows the next role, timeline & salary band — growth without leaving.

3

AI Attrition Risk Prediction

Flags flight-risk employees early using engagement & performance signals.

4

Knowledge Capture Repository

Continuous, lightweight documentation of project & tacit knowledge.

5

Bonus & Training-Bond Automation

Tracks milestone-based retention bonuses and bond agreements automatically.

6

Mentorship & Recognition Module

Pairs interns with mentors and gamifies recognition for engagement.

Before vs. After TalentTrack

BEFORE

- Informal, undocumented onboarding
- No visibility into growth or career path
- Reactive response to resignations
- Knowledge lost when someone exits
- Ad hoc, inconsistent retention bonuses



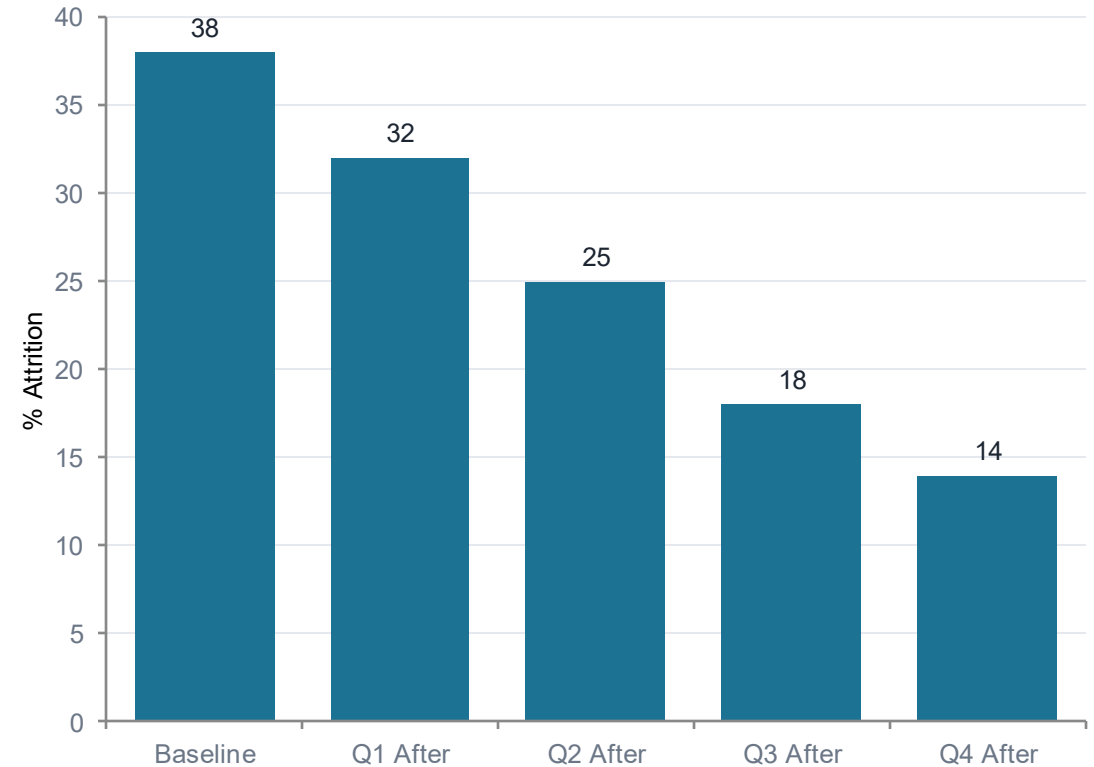
AFTER

- Structured skill tracking from day one
- Transparent, milestone-based career pathway
- Proactive alerts before someone decides to leave
- Continuous knowledge capture & handover
- Automated, consistent bonus/bond management

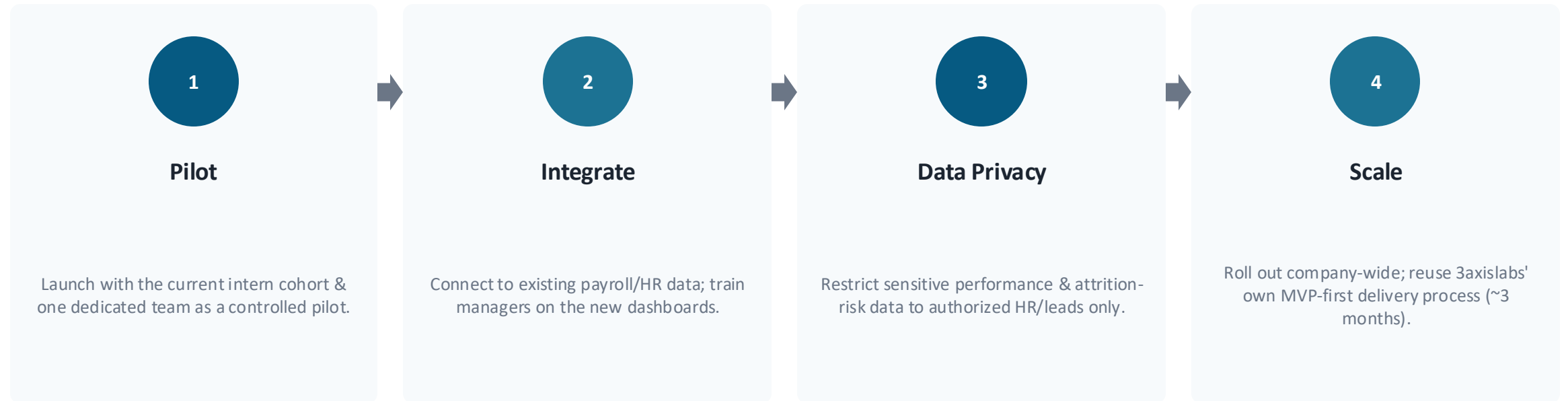
Why This Matters for the Business

- Lower recruitment & re-training costs per role
- Higher intern-to-employee conversion & loyalty
- Preserved institutional knowledge → fewer delivery delays
- Stronger employer brand (SLASSCOM, Yarl IT Hub, GoodFirms visibility)
- Data-driven, fairer compensation & promotion decisions

Expected Attrition Trend (Illustrative Projection)



Rolling TalentTrack Out Responsibly



Because TalentTrack is built on 3axislabs' own existing tech stack (React, Spring Boot, PostgreSQL, Vertex AI), it can realistically follow their proven MVP-first delivery model.

Technology Alone Isn't Enough

1

Pair TalentTrack with Fair Pay Reviews

Benchmark salaries regularly against the market, even on an SME budget.

2

Formalize Training Bonds

Use structured, transparent agreements tied to specific training investments.

3

Build a Visible Promotion Ladder

Publish clear criteria and timelines for intern → junior → senior progression.

4

Strengthen Mentorship Culture

Extend existing team retreats & advisor network into structured mentorship.

5

Review Quarterly via Dashboard

Use TalentTrack data in leadership reviews to catch risk early, every quarter.

Thank You

Questions & Discussion